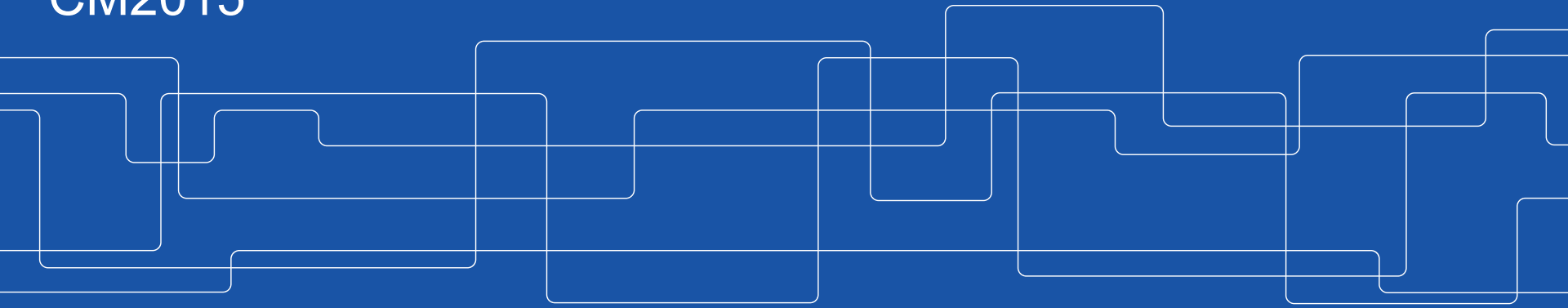
KTH ROYAL INSTITUTE
OF TECHNOLOGY

Project Carrier Course, Part 1

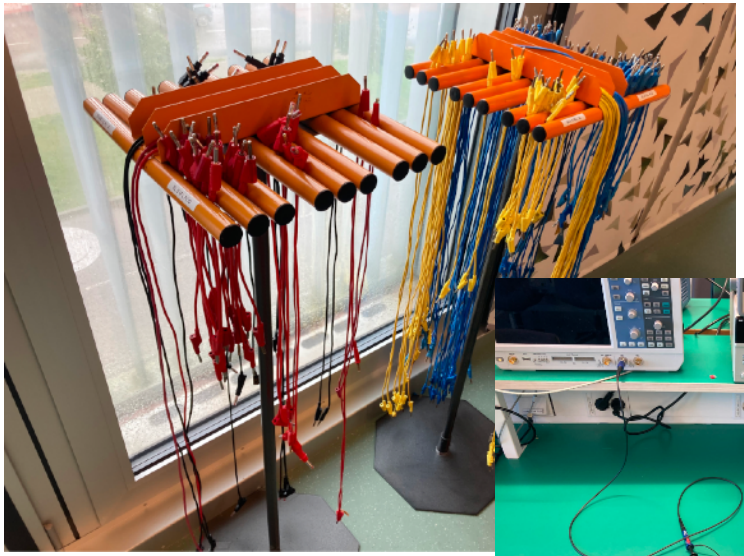
Lecture 2

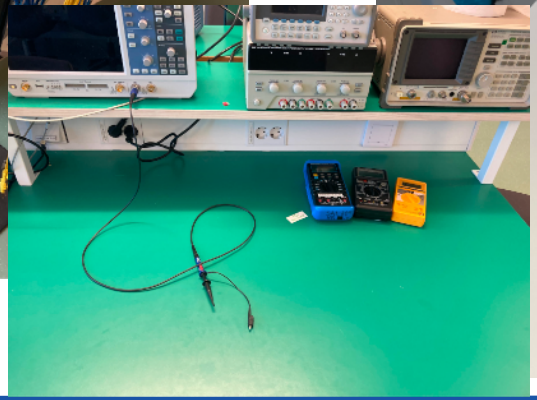
Autumn 2025 - Martin Jacobsson

CM2015

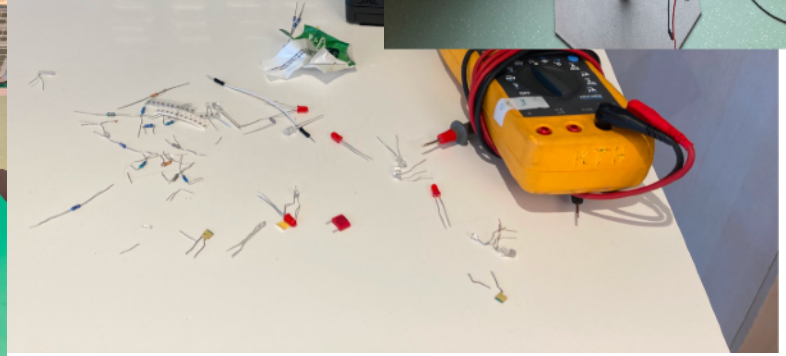


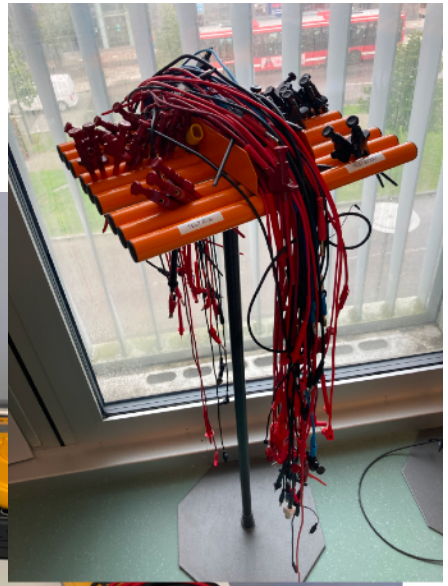
Lab Cleanliness



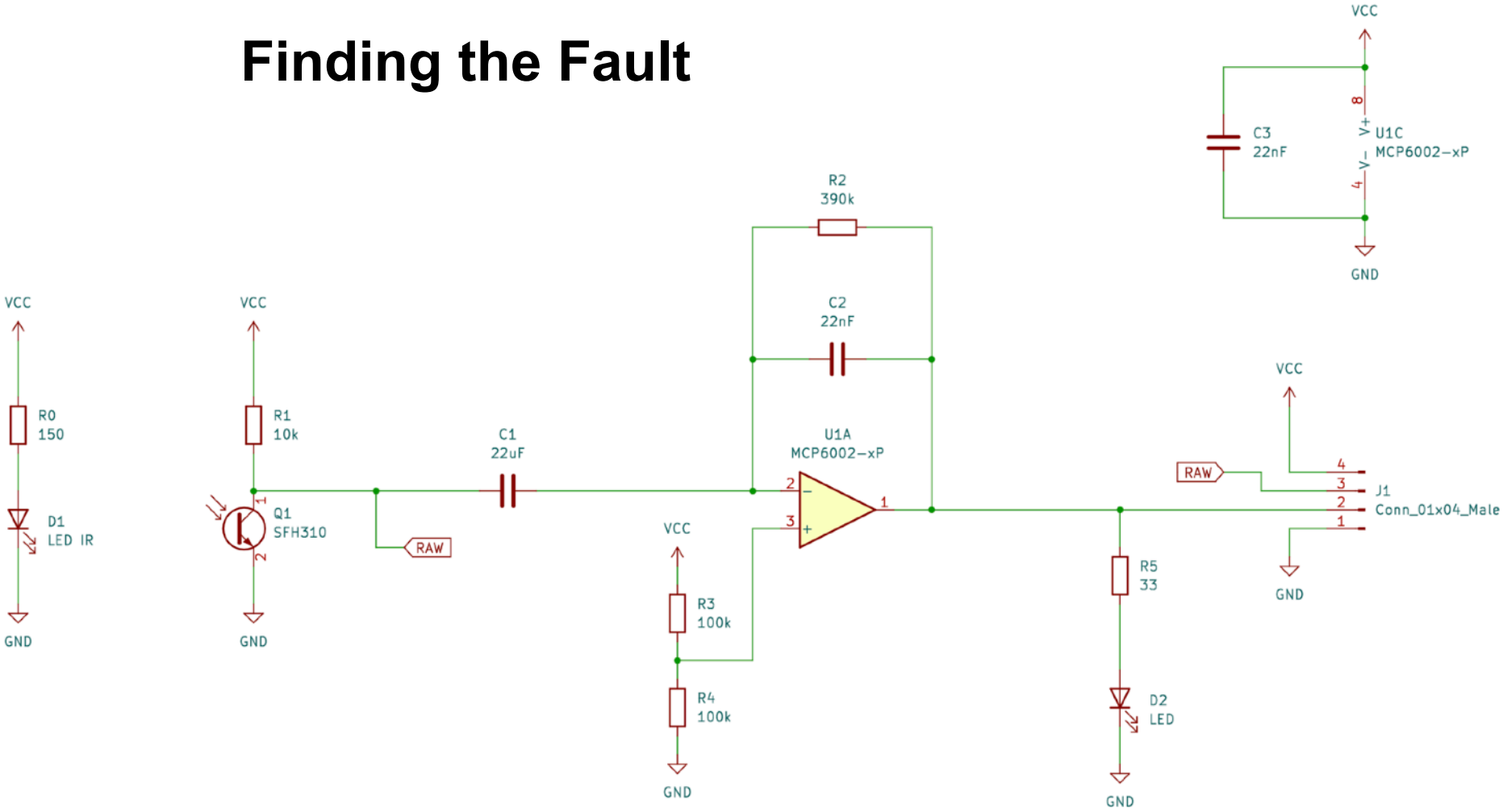




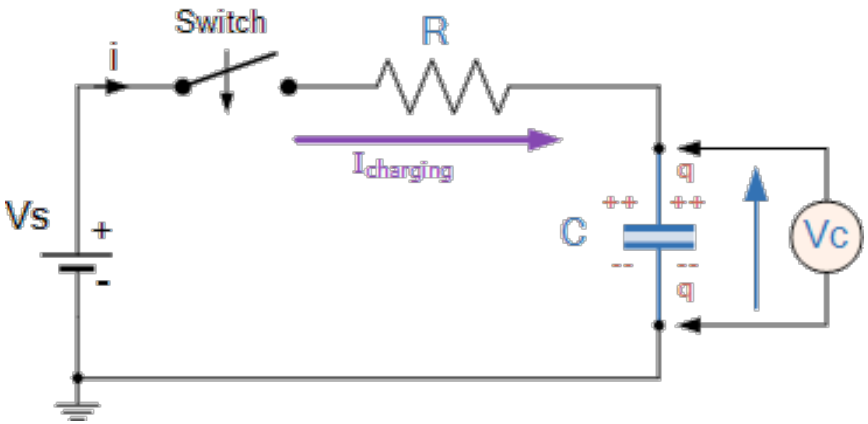




Finding the Fault

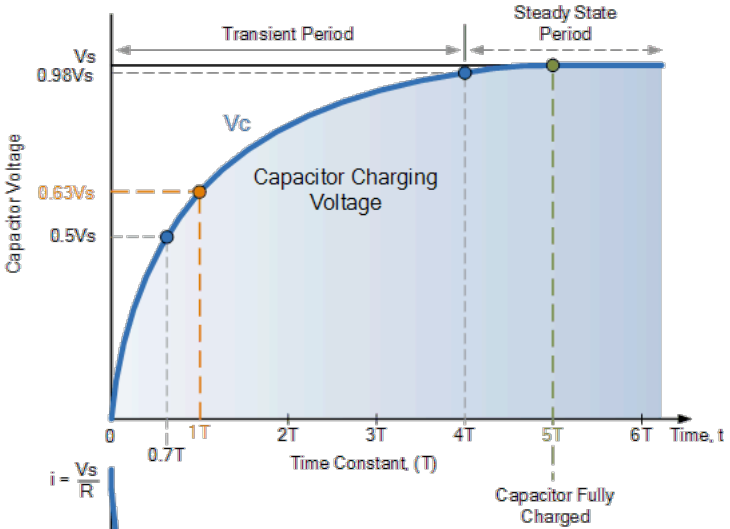


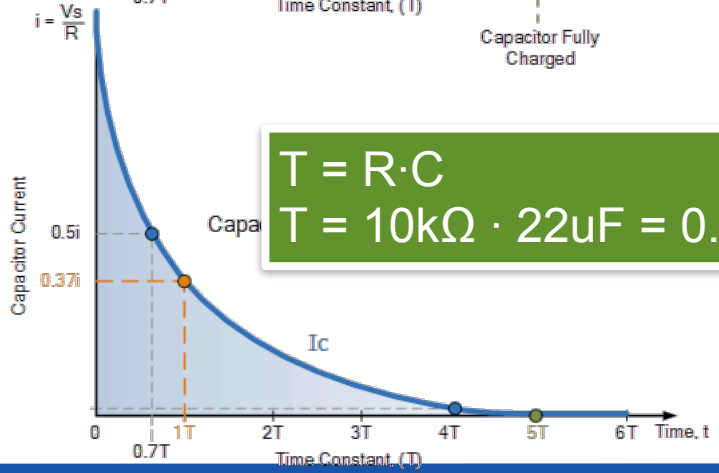
The RC circuit



Ohm's law and Kirchhoff's voltage law:

$V_S - R \cdot i(t) - V_C(t) = 0$



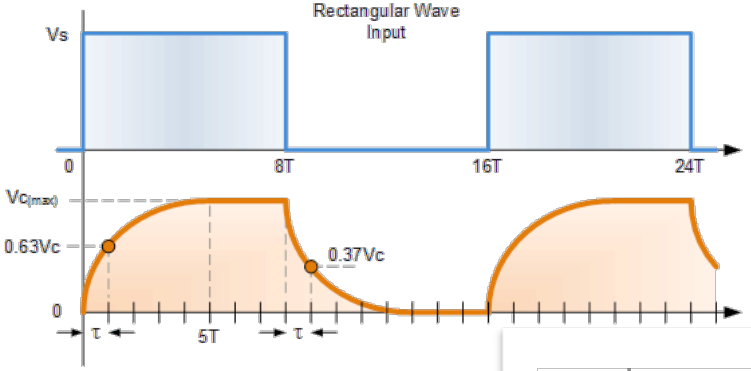


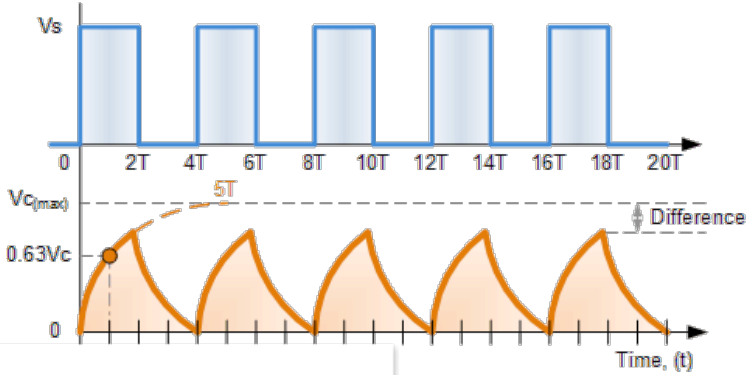
$T = R \cdot C$

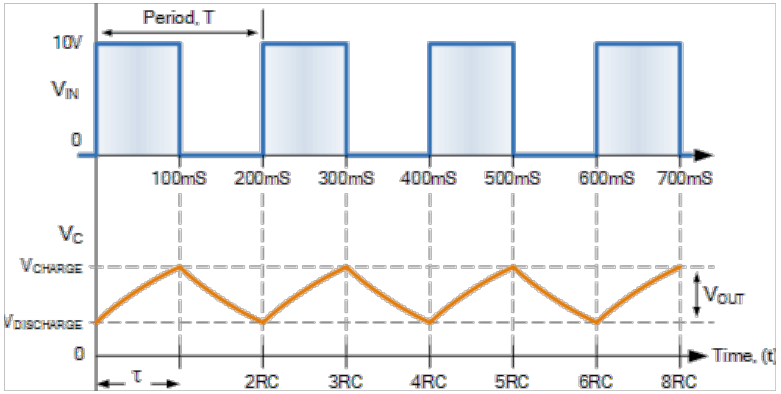
$T = 10k\Omega \cdot 22uF = 0.22\ s$

https://www.electronics-tutorials.ws/rc/rc_1.html

RC-circuit as Low Pass Filter (LPF)









Digital Signal Processing & Filtering

Digital Filtering / Signal Processing

Low Pass Filtering

- Moving Average
- Exponentially Weighted Moving Average
- Median Filter

Digital Amplification:

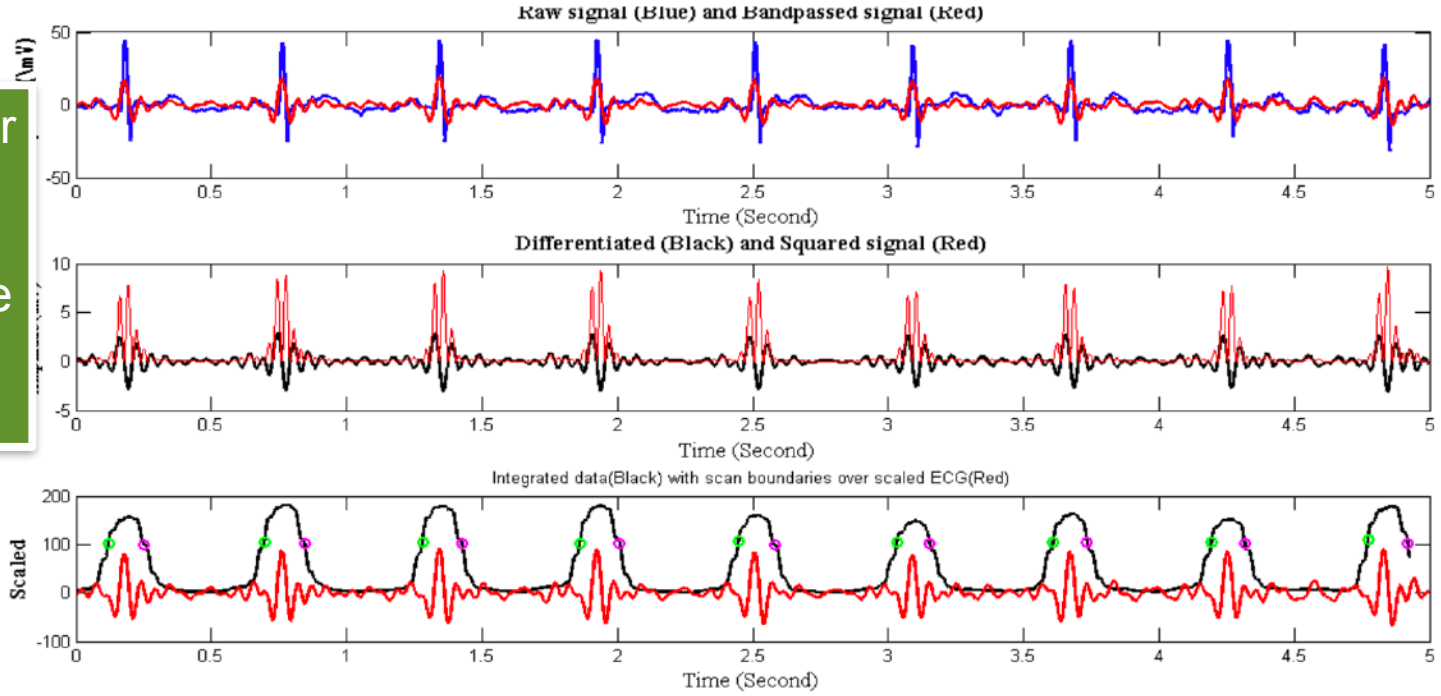
- Linear amplification: 2*signal
- Squaring: signal*signal

Derivative (or differentiation)

Pan, Jiapu; Tompkins, Willis J. (March 1985). "A Real-Time QRS Detection Algorithm". IEEE Transactions on Biomedical Engineering. BME-32 (3): 230–236. DOI: doi:10.1109/TBME.1985.325532

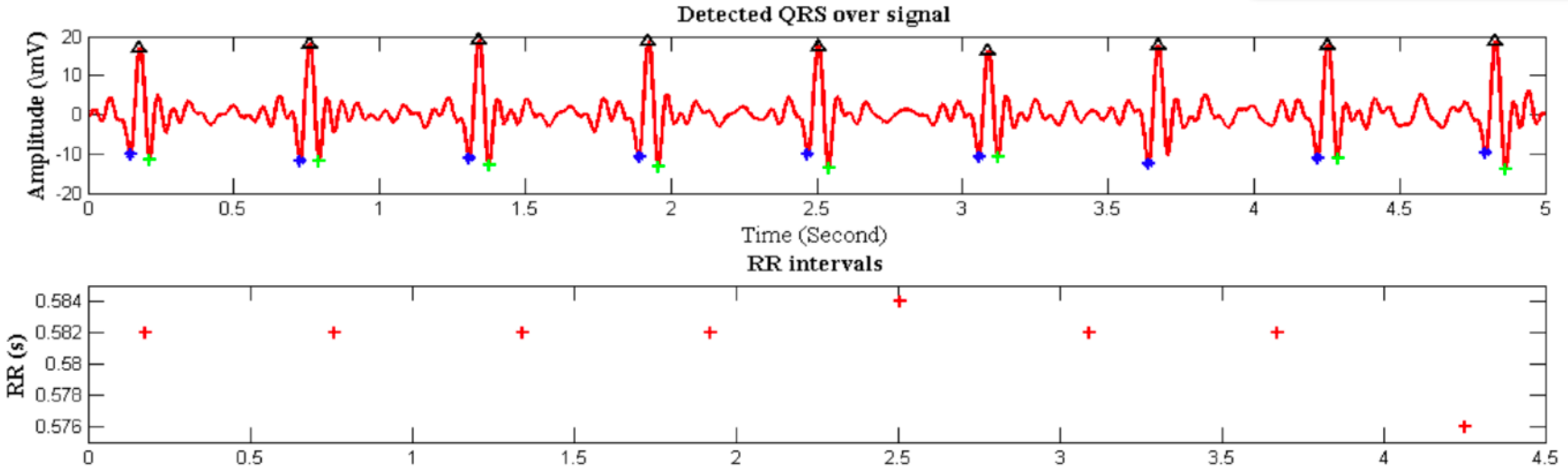
Pan-Tompkins Algorithm (for ECG)

1. Band pass Filter
2. Derivative
3. Squaring
4. Moving Average
5. Fiducial Mark
6. Decision Rules



Pan-Tompkins Algorithm (for ECG)

1. Band pass Filter
2. Derivative
3. Squaring
4. Moving Average
5. Fiducial Mark
6. Decision Rules



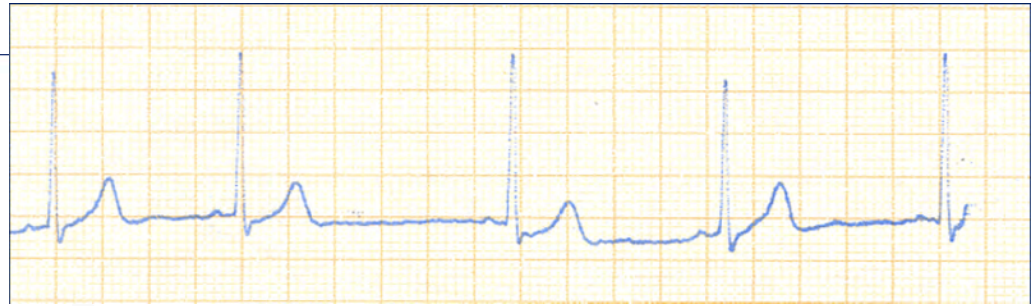
Analog to Digital Converter (ADC)



- Small and fast digital voltage meter
- Measures voltage in an interval (0-3 Volt)
- Outputs an integer value to the software
  - 0 - 1023 (10 bits), 3V ≈ 1023
- Can measure several times per second
  - BBC Micro:bit with software: 400 Hz or faster
- Typical embedded processors have 10 or more ADCs



Analogue or Digital



Analogue

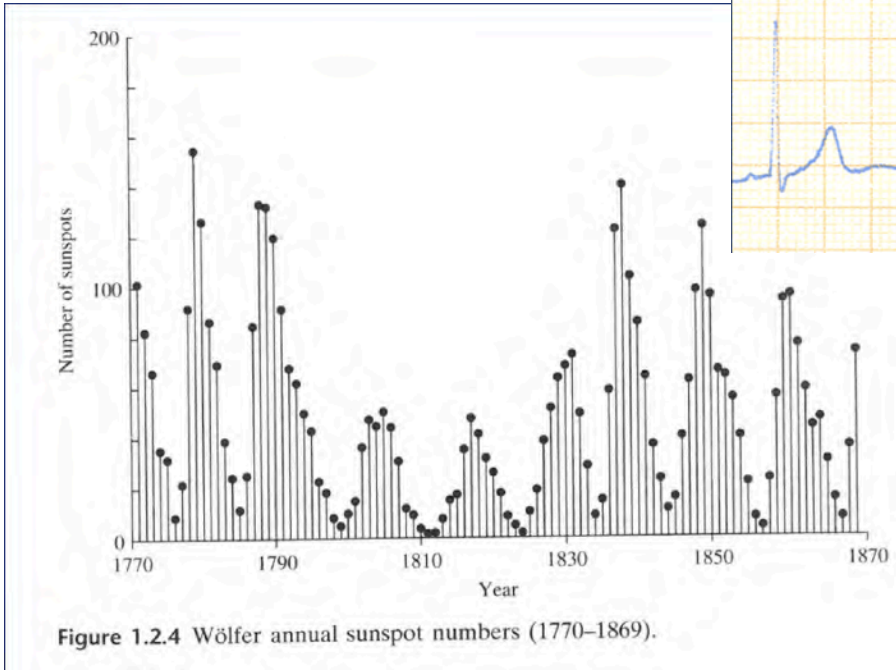
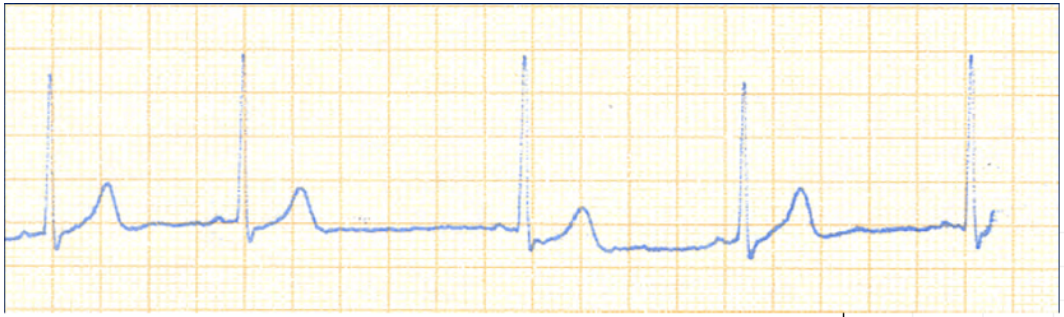


Figure 1.2.4 Wölfer annual sunspot numbers (1770–1869).

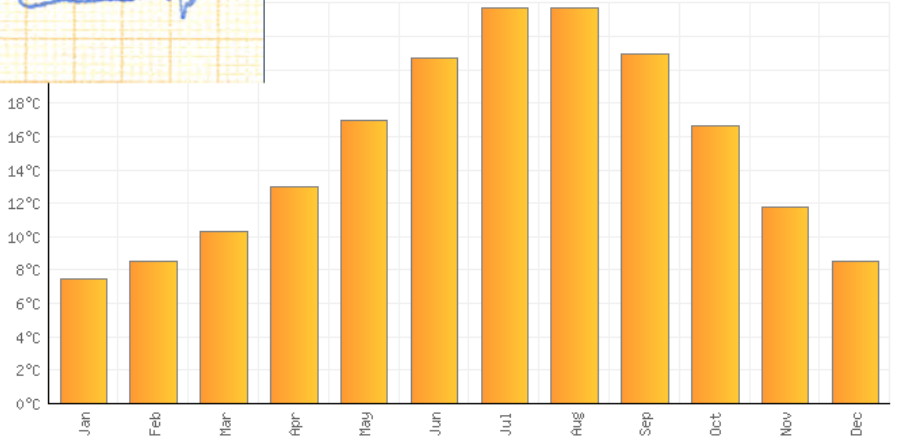
Digital

Time Continuous or Discrete



Continuous

Discrete Time



Average Monthly Temperature (°C) in Rome

Digitalization of analogue/continuous signals

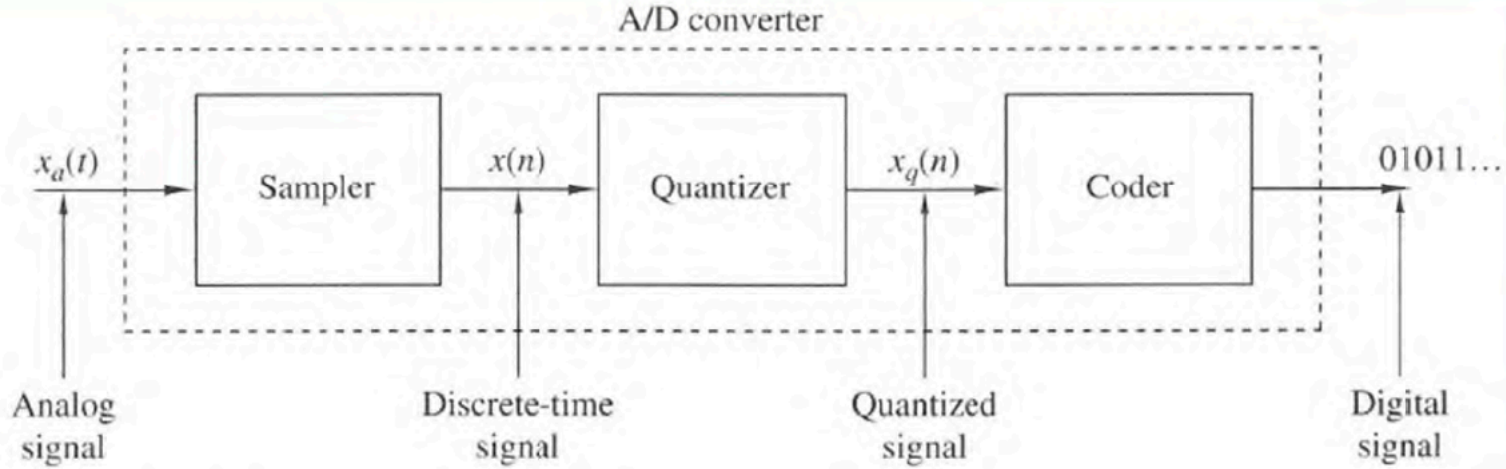


Figure 1.4.1 Basic parts of an analog-to-digital (A/D) converter.

Convert between continuous and discrete

Sampling

T = sampling period
$n \in \mathbb{Z}^+$

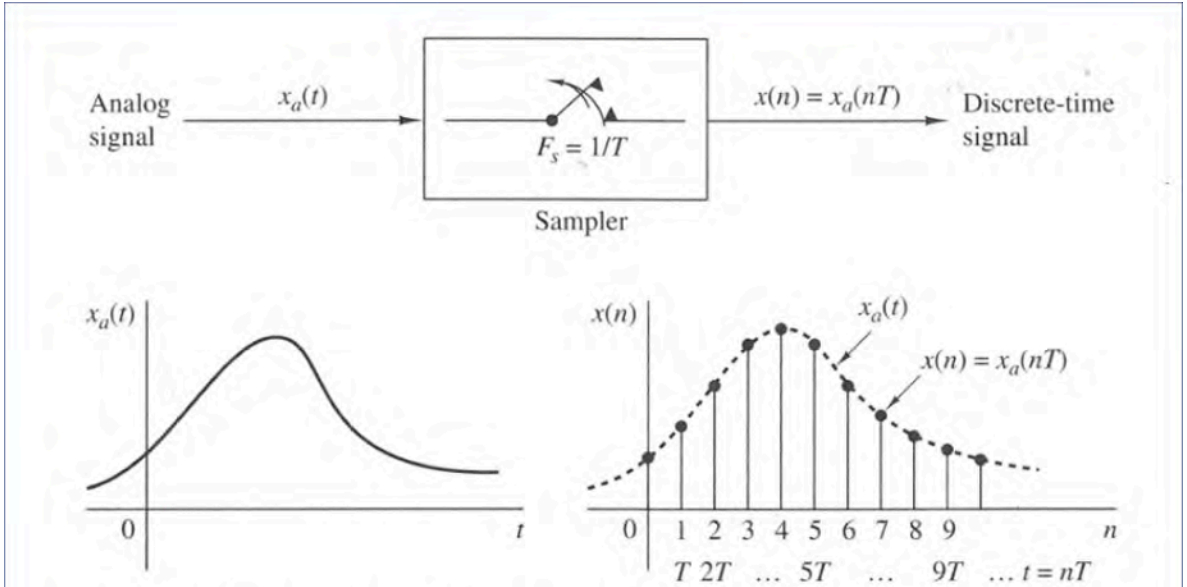
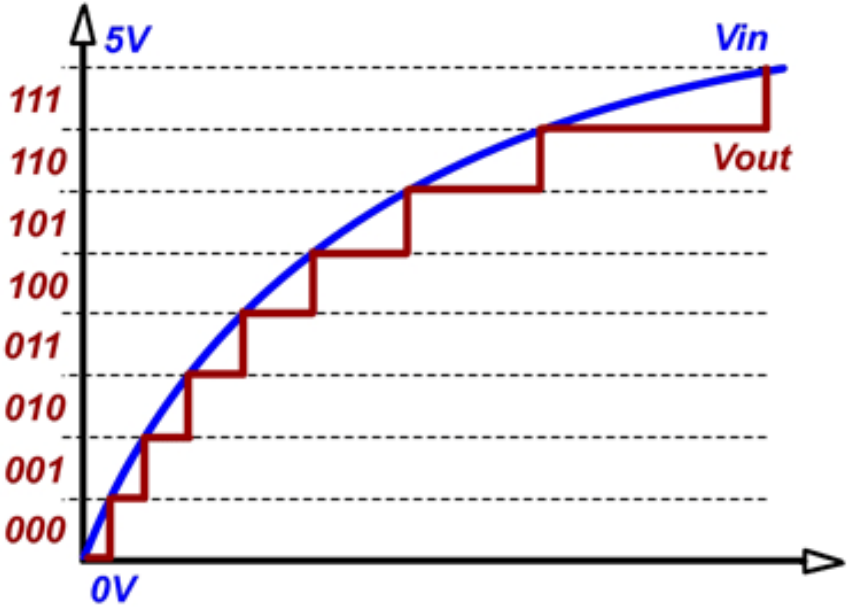


Figure 1.4.3 Periodic sampling of an analog signal.

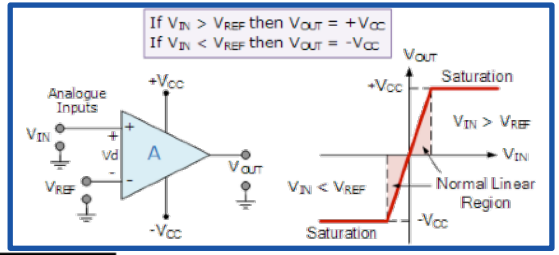
Convert between analogue and digital

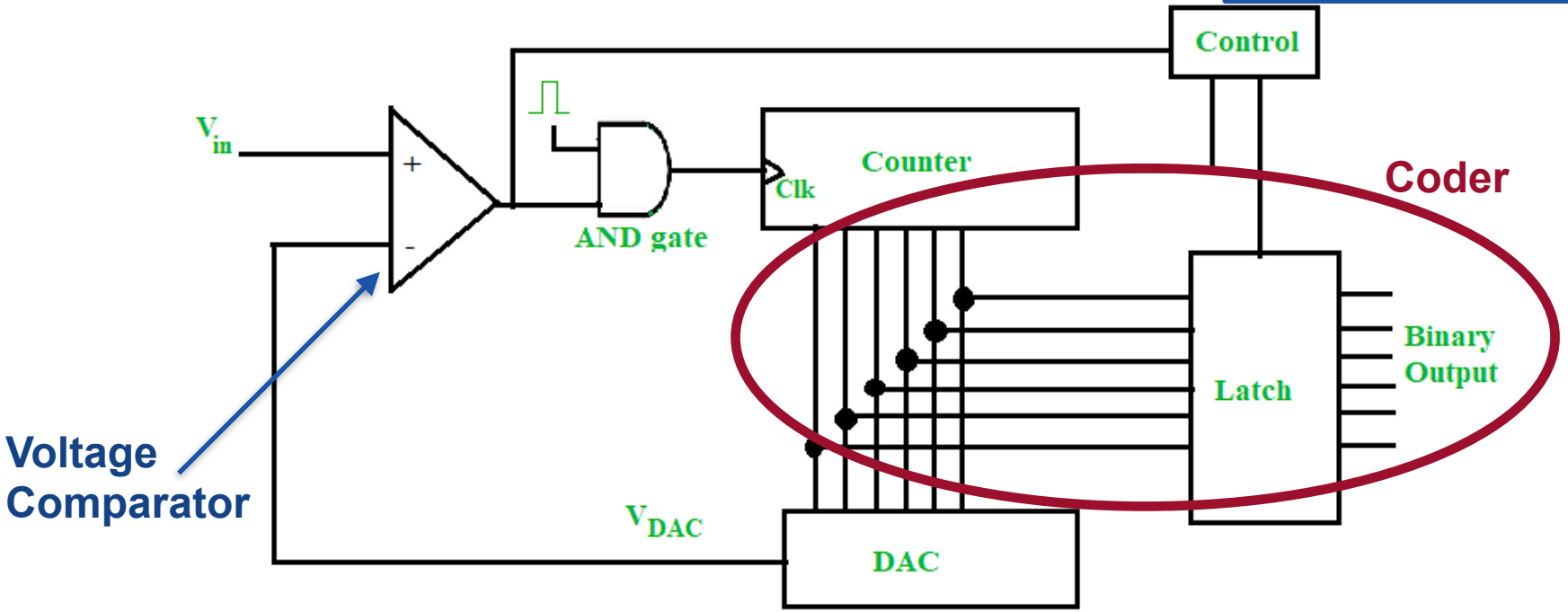
A/D

Quantization



Counter ADC example

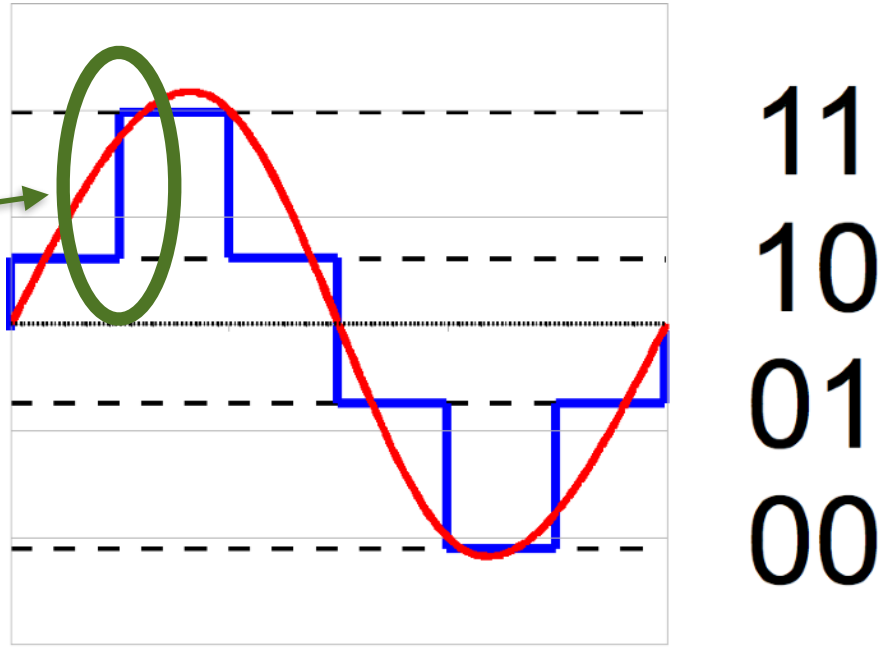




Quantization Error

Amplitude error created by quantisation

The same as adding noise to the signal



Bit depth

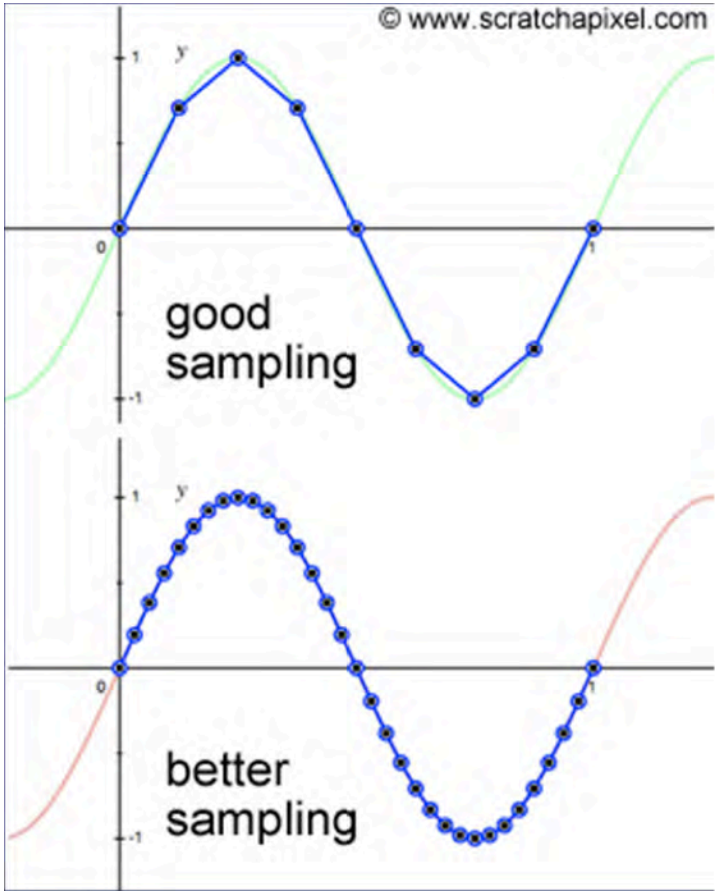
- An 8-bit sampling (quantization) gives a resolution of 256 levels.
- If a signal varies from 0 to 10V, using a 8 bit resolution. Given the sampling value of 3.1415.... V after coding 3.1372 V

- 10/255 = 0.0392 V / level
- 80 * 0.0392 = 3.1372 V

Micro:Bit A/D Converter
Bit depth: 10 bits

Hence, a quantization error of $|3.1415 - 3.1372| = 0.0043$ V

Too low time resolution?



Fourier Series

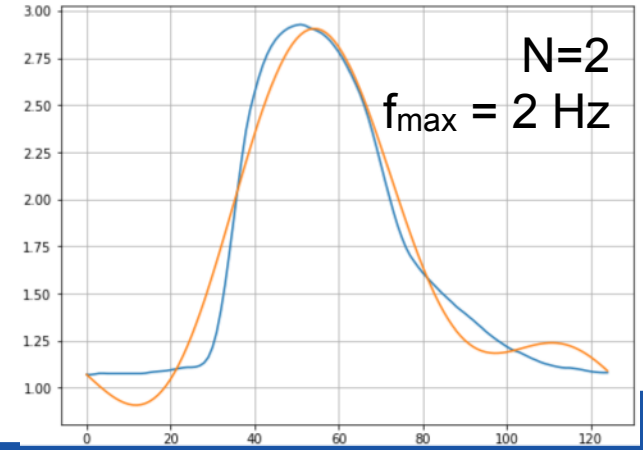
Frequency (rad/s)

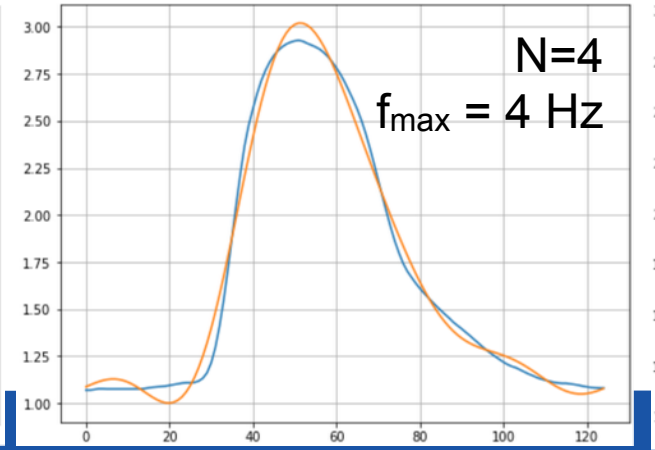
(Almost) any periodic signal can be expressed as a sum of sinusoids

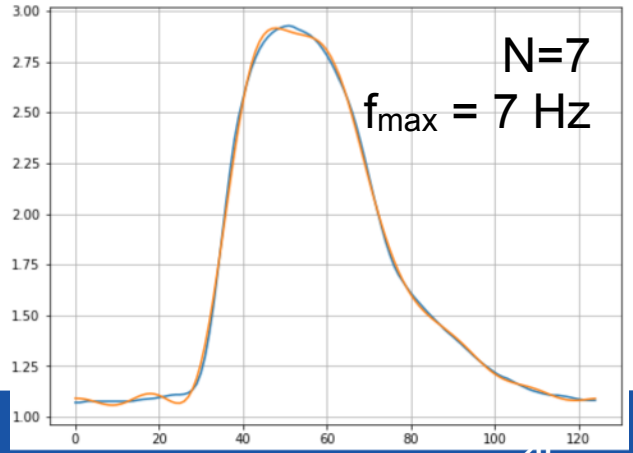
Fourier series, amplitude-phase form

$$s_N(x) = \frac{A_0}{2} + \sum_{n=1}^{N} A_n \cdot \cos\left(\frac{2\pi}{P} n x - \varphi_n\right) \quad \textbf{(Eq.1)}$$

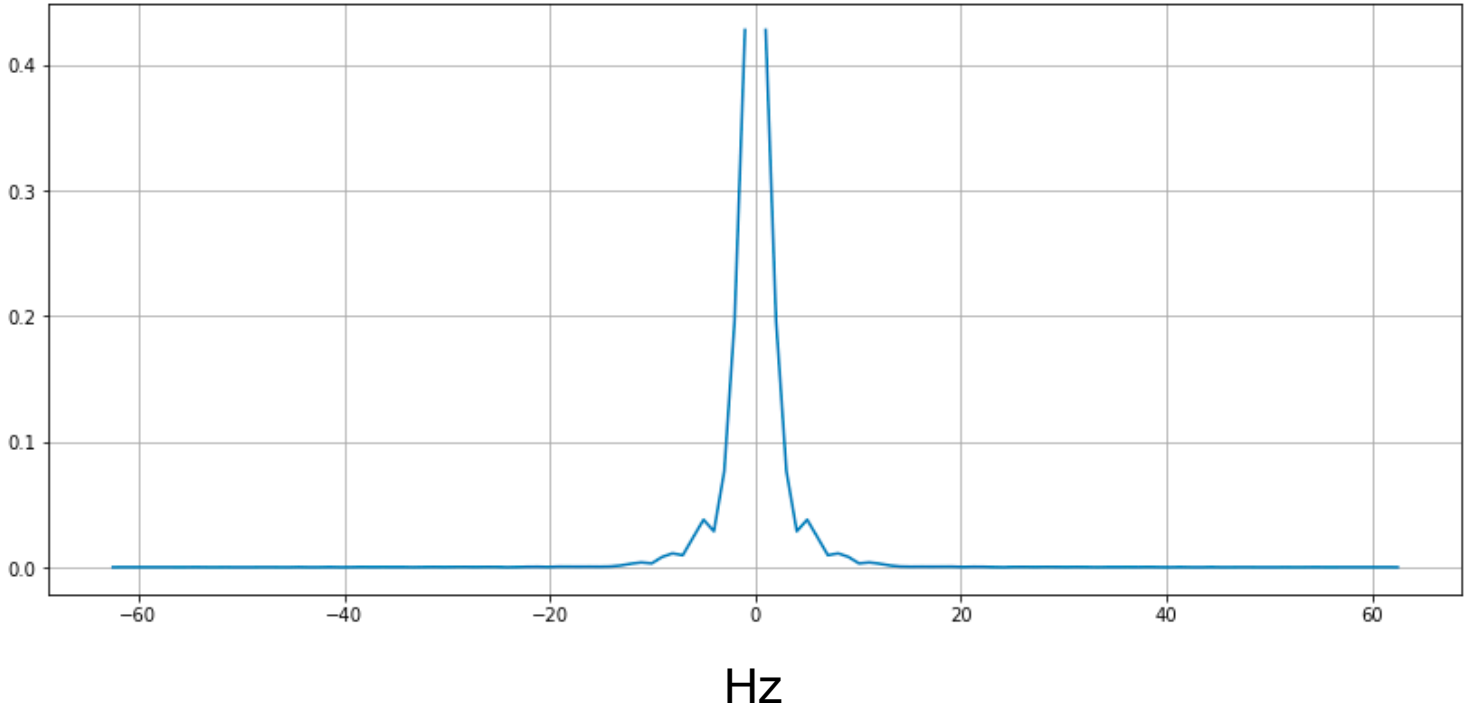
PPG (60bpm)
Blue: original







Discrete Fourier Transform of PPG



Sampling rate

Micro:Bit A/D Converter
Max sampling rate: 400 Hz
with the software environment

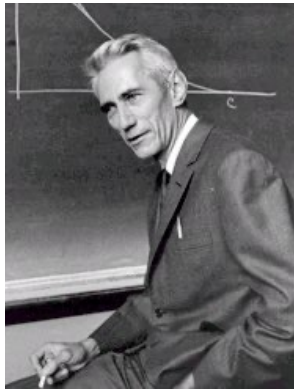
Nyquist-Shannon theorem

- The sampling rate has to be at least twice as fast as the fastest **change**.
- Hence, sampling frequency $f_s \geq 2f_{max}$
- Otherwise the signal will be distorted (aliasing)
- Example:

  Audible sound is in the range of 20 Hz - 20 kHz
  Hence, 40 kHz sampling or more is needed for full sound quality. *A CD uses 44.1 kHz.*

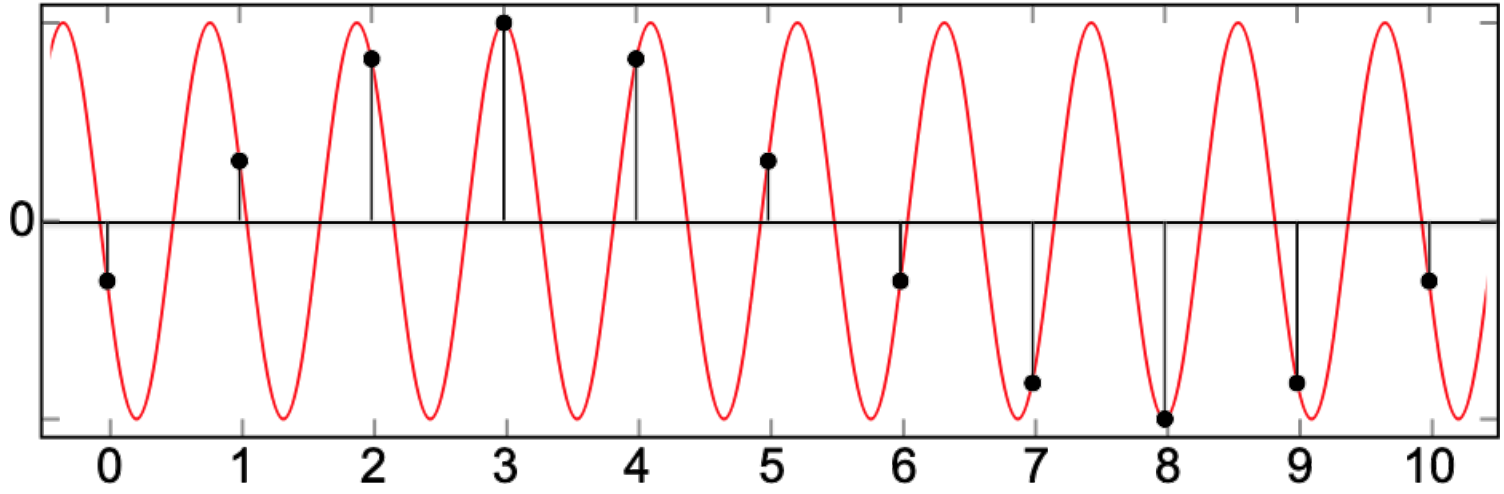


Harry Nyquist



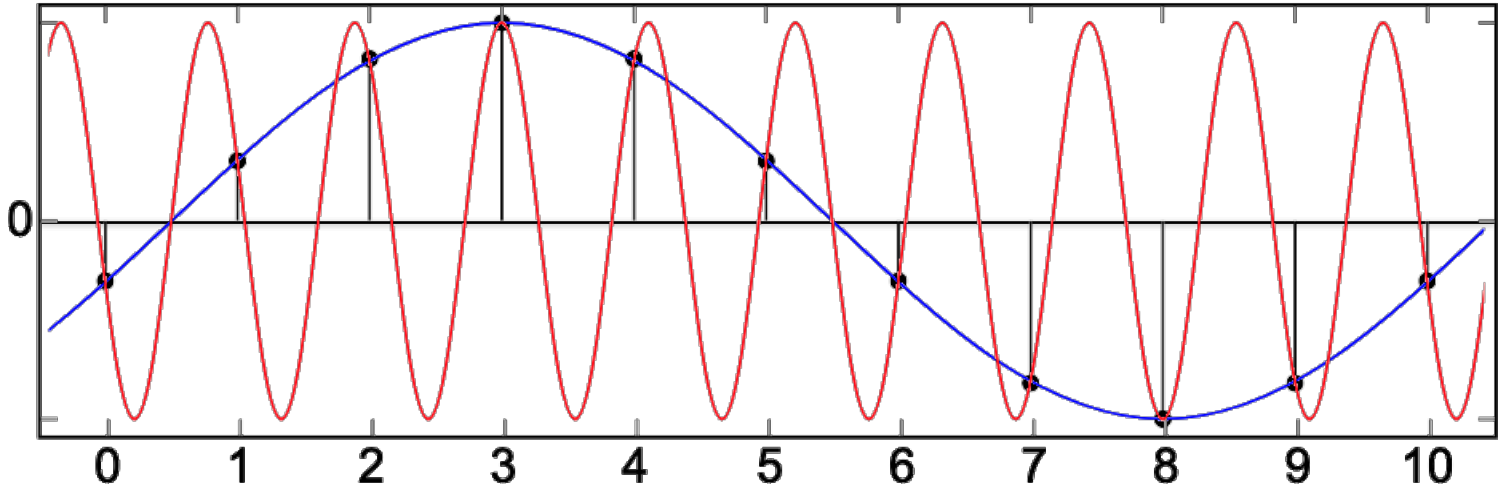
Claude Shannon

Aliasing - One effect of too low sampling



f_1 = 0.9 Hz f_s = 1 Hz

Aliasing - One effect of too low sampling



$f_1 = 0.9$ Hz $f_s = 1$ Hz $f_2 = 0.1$ Hz

To avoid aliasing: Use an analogue low-pass filter before sampling

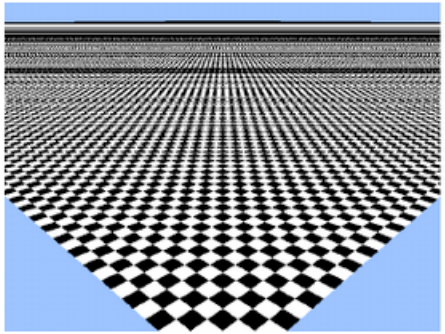
Aliasing in images / video

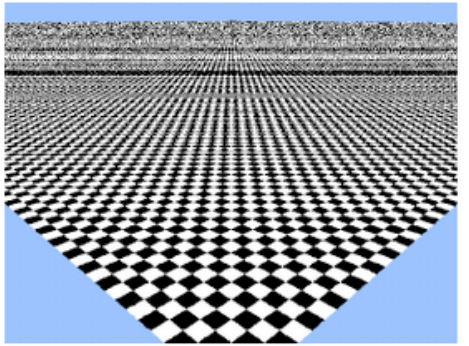
Why do car wheels look like they are spinning backwards at high speed?

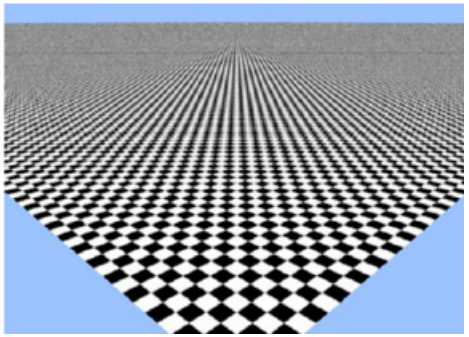
BY HOW IT WORKS TEAM · 04/10/2010



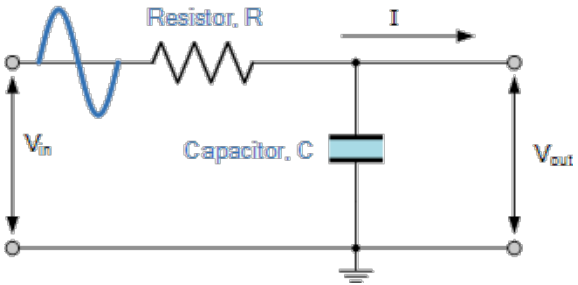
This is due to an effect known as 'aliasing' and is most likely to be observed on TV due to the frame rate of the camera filming it. Video cameras work by capturing lots of still images in a very short space of time. For example, television cameras capture roughly 50 frames a second. This is quite sufficient to fool our eyes and brain into thinking we are seeing a continuous moving image

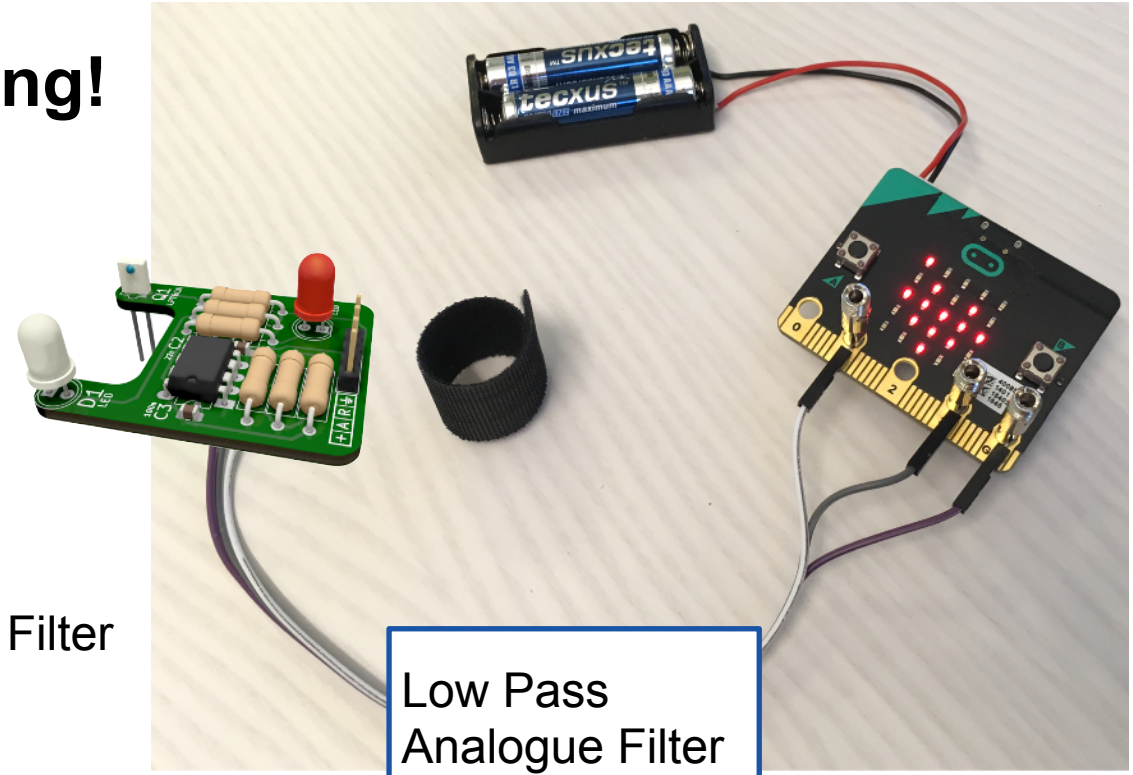






Avoid Aliasing!





First Order Passive Low Pass Filter

We need to select C and R
so that cut-off freq. is less than $f_s/2$